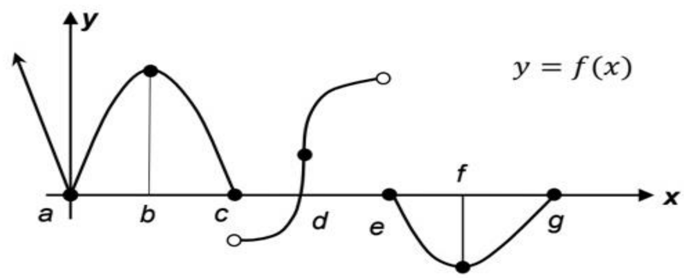
Math-1151 Class Test 03 Section A

1. Consider the function $(x) = x^3$; $x_0 = -1$ and $x_1 = 0$.

   (a). Using definition find the instantaneous rate change of function at x_0. [2]
   (b). Find the average rate of change of function in the interval $[x_0, x_1]$. [1]
   (c). Find the equation of the secant line of $f(x)$ within the specified interval. [1]

2. The graph of function $f(x)$ is given below. State with explanation the locations (value of x) at which $f(x)$ is not differentiable. [2]



3. Find $\frac{dw}{dt}$, if $w = \tan\left(2\sqrt{1+e^t}\right)$. [2]

4. Find $\frac{dw}{dt}$, where $w = y^2$, $y = \sin^{-1} x$, $x = \ln t$. [2]